PALEOMAGNETIC STUDY OF THE UPPER CRETACEOUS IZUMI STRIKE-SLIP BASIN ALONG THE MEDIAN TECTONIC LINE IN SOUTHWEST JAPAN

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Abstract. Along the Median Tectonic Line in southwest Japan, there is a narrow and elongate sedimentary basin of the Upper Cretaceous age, called the Izumi basin, which extends almost 300 km from northwestern Shikoku to western Kii Peninsula. The strata are folded into a syncline that dips approximately 30° to the east. Eastward migration of the Izumi depocenter resulted in a cumulative thickness over 50 km of Late Cretaceous sediments. The structural and depositional patterns are quite similar to other examples of strike-slip basins and suggest that the Izumi basin was formed by a left-lateral strike-slip motion along the Median Tectonic Line. Paleomagnetic studies were carried out in the succession of strata from the lowest horizons in northwestern Shikoku to the upper horizons in Awaji Island 200 km to the east. A total of 730 samples were taken from layers of fine-grained sandstone and mudstone at 94 stratigraphic levels, covering almost half of the succession. Alternating-field and thermal demagnetizations show that most of the specimens have a single and stable magnetization component carried by magnetite. One reversed polarity zone was identified at the lowest portion of the section, and two short reversed zones were noted in the upper portion. Mean directions of the normal and reversed polarities are I=50°, D=76° and I=-47°, D=268°. The mean pole position is 25°N, 203°E. The declinations concentrated to the northeast-southwest is owing to Neogene clockwise rotation of southwest Japan through 50°-60°. The inclinations are compatible with the present latitude of the basin. The pattern of magnetic polarity zones matches a geomagnetic polarity change during the early Campanian to earliest Maastrichtian in accord with previous biostratigraphic studies. Sedimentation rates based on the polarity zonation are 0.7 to 1 cm per year. The extremely high sedimentation rate, as well as other structural characteristics, suggests that the formation of the basin was controlled by extension and a left-lateral motion along the Median Tectonic Line. Despite the unique and complicated structure of the basin, the thick sedimentary pile provides a continuous record of geomagnetic polarity changes. The Izumi basin is an important element in the collision-accretion tectonics of the Japanese arc since the Late Cretaceous.

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Introduction

Recent geologic reinvestigations of the Japanese islands [e.g., Taira et al., 1983] in the context of "collision-accretion" tectonics [Jones et al., 1982; Nur and Ben-Avraham, 1983], have revealed that the Japanese arc is composed of a few tens of geotectonic zones which are grouped into several major terranes. Although the classification of the terranes is still debated by geologists, most agree that the geotectonic zones are bounded by strike-slip fault systems [Ichikawa et al., 1972; Taira et al., 1981]. Among these boundary faults, the Median Tectonic Line (MTL) in southwest Japan is one of the most striking, extending over 500 km along the arc. The dominant sense of motion of the MTL is inferred to be left-lateral since the Cretaceous with a total displacement of several tens of kilometers [Ichikawa, 1980]. Being a major fault system, the MTL belongs to the strike-slip fault network that cuts the eastern margin of Asia [Otsuki, 1985; Faure and Lalevee, 1987]. To the north of the MTL, there is an elongate sedimentary basin containing a thick sequence of Upper Cretaceous deposits, called the Izumi Group. The close similarity in structural and depositional patterns of the Izumi basin to other strike-slip basins, such as the transcurrent San Andreas fault system in western North America [Crowell, 1974], led Taira et al. [1983] to interpret the Iaumi basin as a pull-apart structure created by sinistral motion of the MTL.

Although the unusually thick and continuous succession of the Izumi basin has long been a subject of geological interest, the lack of age-diagnostic fossils prevented correlation to the standard geological time-scale. We conducted

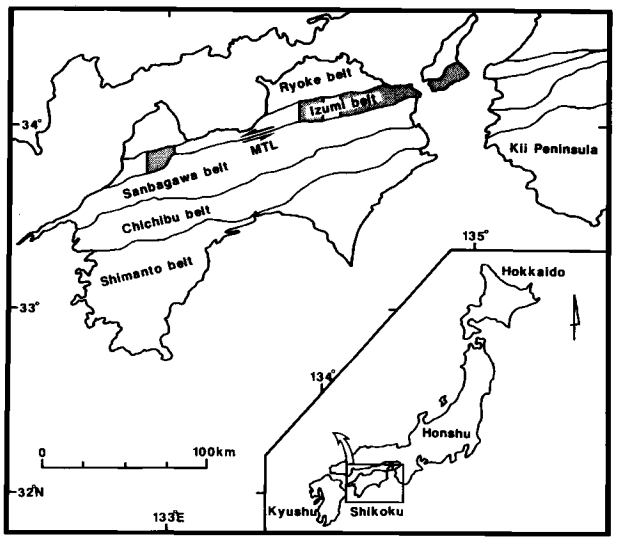


Fig. 1. Geotectonic divisions of Shikoku and western Kii Peninsula in southwest Japan. The shaded zones in the Izumi basin are the areas where intensive sampling was carried out in this study. MTL: Median Tectonic Line.

paleomagnetic studies of the basin succession with the following aims: (1) to construct a magnetic stratigraphy of the Cretaceous deposits, which have a continuity rarely found in Japan, (2) to obtain a reliable Cretaceous paleomagnetic reference pole in southwest Japan. Such a pole would be useful for discussing the relative movements of terranes which are suspected to have been accreted to the Japanese arc during subduction of an oceanic plate, (3) to clarify the displacement of the MTL, which may have exerted tectonic control on the evolution of the basin.

Previous paleomagnetic studies of limited extent in the basin suggested paleomagnetic evidence for left-lateral displacement of the MTL [Kodama, 1986], and found a reversed polarity zone in the Upper Cretaceous, lowermost portion of the Izumi Group [Kodama, 1987]. This article summarizes the previous paleomagnetic results and presents new results, with emphasis on the magnetic stratigraphy. Moreover, we discuss the role of the Izumi basin as an important element of the tectonics of southwest Japan since the Upper Cretaceous.

Geological Setting

In contrast with the highly deformed rocks in the Shimanto Belt (Cretaceous to Tertiary subduction complex) [Taira et al., 1982] and the Chichibu Belt (Jurassic subduction complex) [Taira et al., 1983] which are south of the MTL (Figure 1), the clastic deposits of the Izumi basin are relatively simple in structure. The sedimentary rocks of the Izumi Group occupy a 5-10 km wide basin that extends for approximately 300 km from northwestern Shikoku to Kii Peninsula. Most of the sediments succession consists of a turbidite facies of sandstone and shale interbedded with thin acidic tuff layers. The succession lies unconformably on metamorphic rocks and granite of the Ryoke metamorphic Belt to the north, and contacts the Sanbagawa metamorphic rocks along the MTL to the south. Basal conglomerates and coarse-grained sandstone, rich in rock fragments transported from the Ryoke Belt, are distributed widely along the northern margin of the basin. The Izumi Group contains no detritus from the Sanbagawa Belt [Kanmera and Hashimoto, 1980]. The axial part of the basin is underlain by interbedded sandstone and mudstone, which grade northward into finely bedded and massive layers of mudstone. Acidic tuff layers, which can be traced as marker beds, are interbedded with the turbidites [Suyari, 1973]. Cretaceous volcano-plutonic activity, which was widespread in the inner belt of southwest Japan [e.g., Murakami, 1979], is the most likely origin of these tuff beds.

Unlike other terranes in southwest Japan, the Izumi basin has a somewhat unique and simple synclinal structure. The axis of folding is almost parallel to the MTL and dips several tens of degrees to the east throughout the basin. The structural pattern suggests eastward migration of the depocenter resulted in a cumulative thickness over 50 km [Taira et al., 1983; Nishimura, 1984]. Although occurrences of fossils are quite rare in the basin, preliminary age determinations using ammonites and bivalves [Noda and Tashiro, 1973; Matsumoto and Morozumi, 1980; Bando and Hashimoto, 1984; Morozumi, 1985], and radio-larians [Okamura et al., 1984; Suyari and Hashimoto, 1985] suggest ages ranging from the Campanian to early Maastrichtian. Furthermore, extensive studies of radiolarian and foraminiferan assemblages [Yamasaki, 1987] demonstrate that the age of the strata tends to be younger toward the east, in accordance with the structural pattern of the basin.

The basin shows some unique characteristics in structure and depositional pattern. First, the structure of the southern limb of the fold near the MTL is dominated by en echelon folding suggestive of the left-lateral displacement of the MTL [Ichikawa, 1980], whereas that of the northern limb shows a southwestward-dipping planer structure. The motion of the MTL is indicated also by paleomagnetic measurements [Kodama, 1986] which show that the strata near the MTL underwent counterclockwise rotation about the vertical axis relative to the northern limb, caused by the sinistral drag along the MTL. Second, the paleocurrent axis is directed from northeast-east to southwest-west, suggesting that

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the clastic deposits were fed dominantly from the east to west direction [Suyari, 1973].

Sampling and Laboratory Procedures

Localities of sampling for paleomagnetic measurements are distributed along the 100 km length of the Izumi basin. The areas where intensive sampling was carried out are shown in Figure 1. The areas are the western end of the basin, the middle of the basin in the western and central Asan mountains, the eastern end in northeastern Shikoku, and the south of Awaji Island. Although the lack of outcrops or deep weathering of rocks did not allow complete sampling of the basin succession, we covered about 50 %. The number of sampling localities visited so far amounts to 94 in total, from which 730 specimens were prepared for magnetic measurements.

The westernmost area is underlain by alternating layers of sandstone and shale, folded into a syncline with an ENE-WSW directed horizontal axis. In this area the MTL appears to bend abruptly southward [Takahashi, 1986]. Limited occurrences of megafossils near the basement suggest a Lower Campanian age [Noda and Tashiro, 1973], and radiolarians of the Campanian age have been found in the axial part [Okamura et al., 1984]. Samples for paleomagnetic study were taken from fine-grained sandstone and shale at 17 localities. Because fresh outcrops suitable for paleomagnetic sampling ended, we moved 60 km eastward to the western Asan mountains where 16 localities including two acidic tuff layers were sampled. Further to the east, in the central Asan mountains, is the next sampling area with 31 localities in total. Synclinal folds with eastward-dipping axes are typically seen in those Asan areas and on Awaji Island to the east (Figure 2). Measurements to estimate the plunge of the fold axis suggest an optimum value of 30° eastward [Kodama, 1986]. Most of the samples were taken from fine-grained sandstone and mudstone. Biostratigraphic studies using ammonites [Bando and Hashimoto, 1984] and radiolarians [Yamasaki, 1987; M. Nozoe, unpublished manuscript, 1985] suggest an Upper Campanian age for the succession. The easternmost area is the southern part of Awaji Island where only the northern limb of the syncline crops out. The MTL runs offshore to the south. Samples were collected from 21 different horizons, mainly from mudstones on the eastern part of the island. According to the radiolarian and foraminiferan age determinations by Yamasaki [1987], combined with the study of ammonites [Morozumi, 1985], the age of the succession in Awaji Island can be assigned to the latest Campanian to early Maastrichtian. In addition to those localities, there are nine more localities, each being separated a few kilometers, between Awaji Island and the central Asan mountains.

At each sampling locality, two to five blocks individually oriented with a magnetic compass were collected. The blocks were cored in a laboratory and three to twelve specimens were prepared from each locality. The specimens were measured with a Schonstedt SSM-2A spinner magnetometer or a SCT cryogenic magnetometer. The magnitude of the natural remanent magnetization differs between rock types: 3×10^{-3}

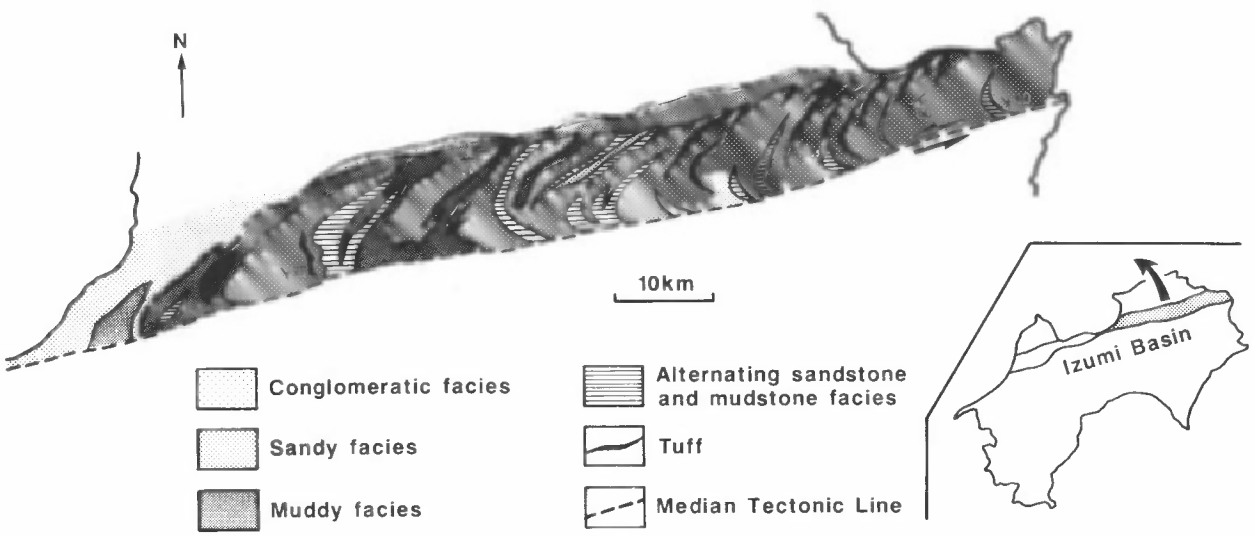


Fig. 2. Geologic map of the Izumi Group in northeastern Shikoku, showing en echelon synclinal structure (redrawn in part after Suyari, 1973).

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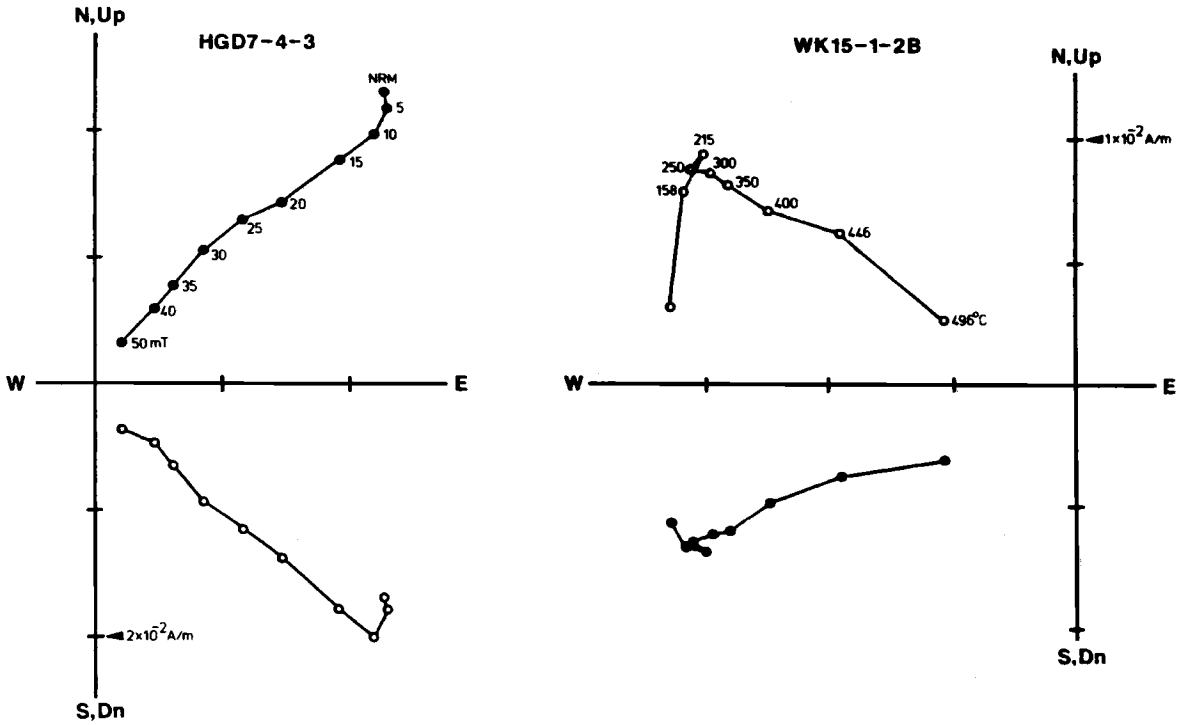


Fig. 3. Examples of vector diagrams; alternating-field demagnetization (left), and thermal demagnetization (right). Solid (open) circles are projections onto horizontal (vertical) plane. Thermal demagnetization was stopped at about 500°C because the thermal alteration was observed above that temperature, which introduced spurious magnetic components even in an ambient field of less than 10 nT.

to 5×10^{-3} A/m for sandstone, and 1×10^{-4} to 4×10^{-3} A/m for mudstone, shale and tuff. Examples of progressive alternating (a. f.) and thermal demagnetization for several pilot specimens are shown in Figure 3, demonstrating stable magnetic behaviors. The low median destructive field intensities (20-25 mT) and unblocking temperatures (<500°C) indicate that magnetite is the magnetic carrier responsible for the remanent magnetization. Therefore, apart from the specimens which had been measured during the previous studies [Kodama, 1986, 1987], those collected in the central Asan mountains to Awaji Island were all subjected to incremental a. f. demagnetization with peak intensities of 10 to 50 mT at an interval of 5 mT. Using the endpoints of the stable magnetization vectors, a characteristic direction for each specimen was determined by a least square estimation.

Paleomagnetic Results

In order to confirm the stability of magnetization, a fold test was applied to a small anticline with approximately 10 m wavelength in the central Asan mountains. A schematic drawing of the fold is illustrated in Figure 4. The magnetization directions from each limb are plotted in Figure 5, which demonstrates the improvement in grouping upon

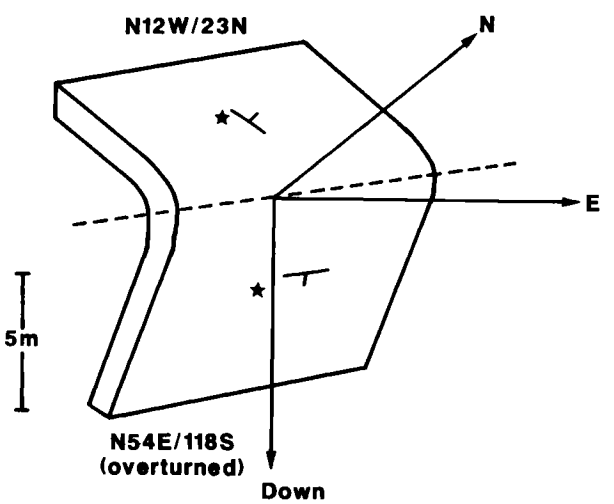


Fig. 4. Schematic drawing of the folded layer subjected to a fold test. Stars are sample sites.

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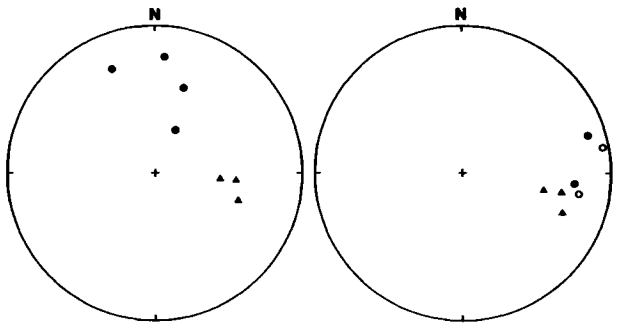


Fig. 5. Plots of magnetization directions before (left) and after (right) correction for bedding tilt. Circles are from the lower limb in Fig. 4 with strike/dip of N54W/118S, and triangles from the upper limb with N12E/23N. Solid (open) symbols in lower (upper) hemisphere.

applying the correction for bedding tilt. The difference in grouping before and after correction can be proven significant by the statistical analysis of McFadden and Jones [1981] at the 99 % confidence level. Although the folding pattern at this locality suggests synsedimentary slumping of soft sediments, the magnetization of which may have been locked sometime after deposition, the positive result of the folding test certifies that the magnetizations were acquired before folding.

Tilt corrections at each locality were performed by taking the plunging axis of the fold into consideration, because the conventional simple unfolding about the strike line produces declination error in some cases [MacDonald, 1980]. The unfolding procedure is, first, to rotate the magnetic directions and poles to bedding through the dip angle of the fold axis, and, second, to rotate each magnetic direction through the dip of the reoriented bed about the strike direction. The convergence in site-mean directions upon applying the bedding corrections in this way is recognized within each limb of the synclinal fold of the basin. According to Kodama [1986], the precision parameters (k) of in situ directions are 40.0 and 3.7 for the northern nine sites and southern seven sites in the western Asan mountains. Those after the

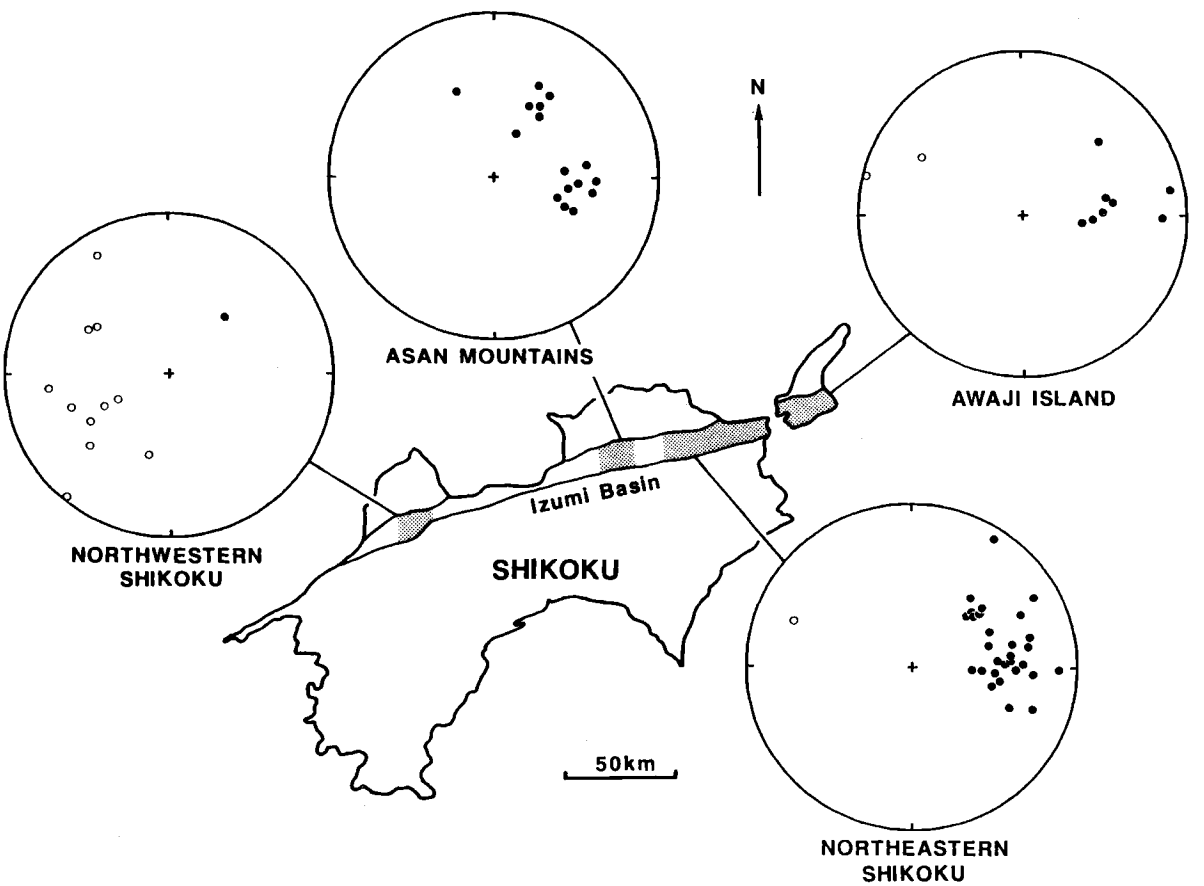


Fig. 6. Plots of site-mean directions divided into four groups according to the study areas. Black: areas sampled; stippled: no paleomagnetic sampling because of poor exposure. Solid (open) symbols in lower (upper) hemisphere.

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TABLE 1. Paleomagnetic Data From the Izumi Group in Shikoku, Southwest Japan

Site	N	Str/Dip	I	D	I_c	D_c	k	α_{95}
WS-1	5	80E/70S	-10.4	215.8	-39.3	251.2	15.8	19.8
WS-3	6	80E/70S	-18.0	227.7	-30.1	263.8	22.5	14.4
WS-4	7	85E/22S	-42.3	217.3	-54.9	242.6	49.0	8.7
WS-5	8	32E/80E	-29.6	159.2	-45.2	238.4	83.7	6.1
WS-6	5	50E/70S	-17.3	193.9	-36.7	227.5	42.9	11.8
WS-7	7	82E/42S	-59.4	242.9	-45.8	300.6	46.2	9.0
WS-8	9	90E/55S	-51.6	228.2	-48.3	304.7	138.8	4.4
WS-9	9	90E/08N	-10.0	332.0	-17.4	331.3	23.9	10.8
WS-12	8	80W/30N	-62.4	306.3	-63.0	243.6	22.3	12.0
WS-13	5	25W/20S	-31.4	205.4	-48.1	194.6	21.4	16.9
WS-14	5	45E/35N	36.6	76.4	50.2	44.2	13.8	21.4
WS-15	7	70W/52N	-47.4	232.3	-1.9	218.9	29.5	11.3
AS-1	12	85E/40S	67.3	60.9	47.9	113.8	105.0	4.3
AS-2	11	45E/30S	57.0	65.9	38.5	93.0	379.7	2.3
AS-3	5	75E/30S	68.6	75.9	51.7	115.5	490.1	3.5
AS-4	18	55E/35S	72.1	46.2	52.6	101.1	49.5	5.0
AS-5	16	68E/40S	53.8	54.8	48.0	96.7	181.4	2.7
AS-6	12	83E/38S	52.2	73.5	40.0	101.3	113.2	4.1
AS-7	11	60E/40S	71.1	2.7	58.0	111.2	311.8	2.6
AS-8	12	55E/30S	52.8	48.1	43.1	84.9	104.5	4.3
AS-9	18	45E/46S	59.5	27.9	55.9	88.3	211.7	2.4
AS-10	16	20W/40N	82.2	325.1	67.2	27.4	344.1	2.0
AS-11	16	30W/30N	73.9	49.1	41.1	35.0	255.2	2.3
AS-12	7	20W/45N	70.7	8.9	52.1	27.0	98.0	6.1
AS-13	12	85W/72S	-50.5	337.7	44.5	337.5	78.4	4.9
AS-14	10	20W/80N	66.4	46.4	49.5	33.8	404.3	2.4
AS-15	5	37W/13N	71.7	17.1	53.7	38.0	23.3	16.2
AS-16	12	30E/90N	65.6	42.3	40.6	26.2	281.3	2.6
ES-1	6	80E/50N	45.3	36.5	7.2	32.6	18.5	16.0
ES-2	8	12E/33S	62.4	10.1	46.5	52.2	12.3	16.5
ES-3	9	40W/50N	65.1	117.5	29.2	80.0	54.1	7.1
ES-4	8	15W/20N	69.8	27.3	52.6	47.9	22.8	11.8
ES-5	6	17W/30N	73.8	109.4	46.7	85.7	25.0	13.6
ES-6	9	78W/38S	40.7	65.7	48.7	94.4	252.3	3.2
ES-7	9	70E/32S	62.2	45.6	55.7	93.2	54.6	7.0
ES-8	9	45E/40S	65.8	45.6	42.1	88.4	61.2	6.6
ES-9	8	40E/40S	62.9	60.7	33.3	88.4	43.5	8.5
ES-10	8	20E/50S	58.2	26.4	27.2	64.4	70.8	6.6
ES-11	7	44E/40S	57.5	71.2	27.2	90.7	11.3	18.8
ES-12	6	40E/50S	48.1	47.9	26.4	76.1	58.8	8.8
ES-13	4	40W/32N	74.9	18.7	44.1	40.8	23.0	19.6
ES-14	5	80E/50S	31.6	85.2	11.8	91.4	5.0	37.8
ES-15	5	70W/30N	66.7	74.2	44.1	50.5	17.4	18.9
ES-16	9	55W/42N	53.8	66.1	16.6	60.0	24.5	10.6
ES-17	6	50E/60S	54.2	338.3	60.2	93.5	10.3	21.9
ES-18	5	35E/77S	44.2	327.5	50.2	75.1	15.1	20.4
ES-19	8	45E/50S	57.1	29.1	37.8	77.1	59.7	7.2
ES-20	6	70E/35S	59.4	63.6	45.0	99.0	57.7	8.9
ES-21	6	50E/50S	48.4	14.6	47.1	66.5	17.6	16.4

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TABLE 1. (continued)

Site	N	Str/Dip	I	D	I_c	D_c	k	α_{95}
ES-22	7	40E/30S	50.4	18.2	49.1	50.3	40.6	9.6
ES-23	8	57E/57S	56.6	16.8	45.4	87.9	152.2	4.5
ES-24	5	44E/44S	67.1	49.7	37.3	91.1	75.5	8.9
ES-25	8	10E/60S	52.1	319.8	49.8	48.3	45.4	8.3
ES-26	7	70E/36S	46.4	57.1	39.7	82.8	14.9	16.2
ES-27	6	74E/59S	59.5	75.4	22.9	109.3	109.2	6.4
ES-28	9	88E/56S	49.3	44.7	48.7	103.9	97.2	5.2
ES-29	12	67E/40S	49.3	53.5	39.7	84.4	64.7	5.4
ES-30	8	50E/46S	41.4	9.2	50.0	51.3	7.6	21.4
ES-31	2	80E/54S	60.3	61.5	36.4	112.1	20410.2	1.7
ES-32	3	34E/23N	12.4	286.6	-38.2	288.2	18.5	29.5
AJ-1	8	54E/31S	56.3	54.0	70.0	104.0	12.6	16.2
AJ-2	5	57E/62S	52.0	344.3	77.6	147.5	39.8	12.3
AJ-4	8	60E/38S	63.2	37.2	73.9	132.0	43.9	8.4
AJ-5	6	68E/46S	54.2	33.9	69.9	125.9	57.5	8.9
AJ-6	5	65E/25S	29.0	36.6	57.8	39.5	10.9	24.2
AJ-7	7	67E/31S	50.8	49.4	72.8	102.5	13.6	16.9
AJ-8	6	75E/29S	58.9	55.2	75.3	145.2	12.7	19.5
AJ-11	6	79E/22S	-46.6	292.7	-47.1	329.5	10.1	22.1
AJ-13	5	29E/56S	-57.6	289.2	-27.2	298.9	60.6	9.9
AJ-17	7	46E/54S	44.6	67.7	32.7	98.2	11.5	18.6

WS: western Shikoku; AS: Asan mountains; ES: eastern Shikoku; AJ: Awaji Island. Data of sites AS-1 to AS-16 are from Kodama [1986]. N: number of specimens. Str/Dip: strike and dip of bedding at sample site. I, D: in situ inclination and declination. I_c, D_c: inclination and declination after correction of bedding attitude and plunging fold axis. k, α_{95}: Fisher's precision parameter and semi-angle of 95% confidence cone.

tilt corrections are raised to 66.6 and 27.6, respectively.

After rejecting the localities which showed either unstable magnetic behavior during demagnetization based on vector plot or poor grouping of directions (k<5), a total of 70 site-mean directions were obtained from the Izumi basin. Table 1 is a summary of the paleomagnetic results. Figure 6 shows the selected directions divided into four groups according to locality. In terms of magnetic polarity, there are 14 sites with reversed polarities: 10 of them are located in the westernmost area and the rest in the northeastern Shikoku and Awaji Island. The mean direction is $I=-46.8^\circ$, $D=267.7^\circ$, $\alpha_{95}=18.6^\circ$. All the others show normal polarities with the mean of $I=50.2^\circ$, $D=76.4^\circ$, $\alpha_{95}=6.1^\circ$. The antipodal ralationship demonstrates that the data set pass a reversed test. The overall mean direction of the 70 sites is $I=49.6^\circ$, $D=78.6^\circ$ with $\alpha_{95}=5.9^\circ$. The mean pole position is 25°N, 203°E. The declinations aligned to the northeast-east or southwest-west are consistent with the paleomagnetic directions reported for the Cretaceous igneous rocks in the inner belt of southwest Japan [Otofuji and Matsuda, 1984] which has been interpreted as being rotated clockwise through 50°-60° by the opening of the Japan Sea in the early Miocene [Otofuji et al., 1986]. The tilt-corrected inclinations are compatible with the present geocentric axial dipole field, suggesting that the deposits accumulated in place and were not displaced in latitude after deposition.

Discussion

Figure 7 is a summary of the paleomagnetic results of the Izumi Group, showing stratigraphic levels sampled and a possible correlation to the geomagnetic polarity time-scale for the Upper Cretaceous [Kent and Gradstein, 1985]. We correlate the reversed polarity zone of approximately 3 km thickness in the lowest part of the section to a part of reversed polarity chron

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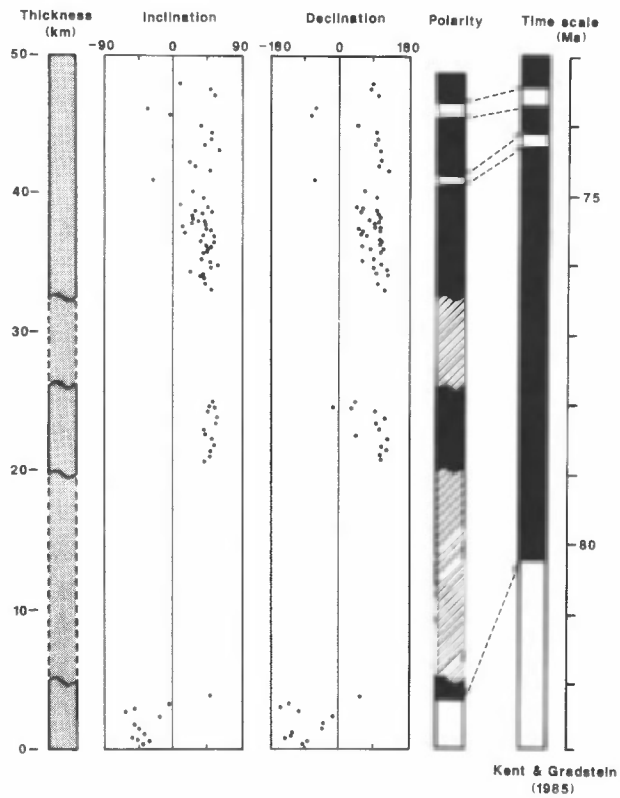


Fig. 7. Paleomagnetic results from the Izumi Group, and correlation of the polarity zones to the Upper Cretaceous polarity time-scale (after Kent and Gradstein, 1985). Black: normal polarity; white: reversed polarity; ruled pattern: no data available.

in the early Campanian. This correlation is consistent with fossil ages of early to middle Campanian from the sedimentary rocks [Noda and Tashiro, 1973; Okamura et al., 1984; Yamasaki, 1987]. Likewise, the normal polarity layers in the middle to upper portions, including one normal polarity layer over the lowest reversed polarity zone, may correlate with a normal polarity chron in the Upper Campanian, although the entire succession has not been fully examined due to the lack of outcrops. Above this thick normal polarity zone, there is a short reversed polarity interval located in the eastern Asan mountains. The uppermost normal polarity zone sonsists of the eight localities in Awaji Island. Two short reversed polarity zones occur above the thick normal zone, and the three localities at the highest stratigraphic level have normal polarities. The two short reversed subzones from the upper structural level are assigned to reversed polarity chrons in the earliest Maastrichtian. Radiolarian and foraminiferan assemblages from these higher horizons, which suggest ages of latest Campanian to early Maastrichtian [Yamasaki, 1987], support our magnetic polarity correlation.

Our interpretation of the magnetic polarity stratigraphy for the Izumi Group provides some constraints for understanding the geologic development of the basin. Correlation of these polarity zones to a geomagnetic polarity time-scale allows us to estimate the average depositional rate of the basin. According to the time-scale by Kent and Gradstein [1985], the age of the polarity boundary at the lowest stratigraphic level is approximately 80 Ma and the next one up is 74 Ma. Within this interval, approximately 40 km thick sediments were accumulated, so the average sedimentation rate for this section is 0.7 cm/year. A similar estimate can be made for the upper part of the section using the polarity boundaries of the two short reversed intervals, resulting in a rate as much as 1 cm/year. From the extremely high sedimentation rate we infer that the sedimentation of the Izumi basin was controlled tectonically. In addition, the lateral change of magnetic polarity zones seems to confirm, as speculated in previous studies [e.g., Taira et al., 1983], that the basin was formed by lateral migration of the depocenter following left-lateral strike-slip motion of the MTL during the early Campanian to early Maastrichtian. Like other strike-slip sedimentary basins in southern California [Crowell, 1974] and Norway [Steel, 1976], a bend in the MTL may have caused a laterally migrating uplift and sag, leaving a thick accumulation of sediments along the fault. The accumulation rate estimated from the magnetostratigraphy, therefore, may be equivalent to the average displacement rate of the MTL. I speculate that each turbidite unit of the basin succession may correspond to a discrete fault movement, or earthquake occurrence on the MTL. If true, the average rate of displacement along the MTL was approximately one meter per 100 years during the Late Cretaceous.

In addition, it is concluded from the present study that the Upper Cretaceous terrigeneous clastic rocks of the Izumi basin are useful for paleomagnetic investigations of exotic rocks distributed in the Chichibu and Shimanto Belts (Figure 1) in the outer zone of southwest Japan [Kodama et al., 1983]. Because there is no stable cratonic terrane that provides a reference polar wandering path for the Japanese arc, the autochthonous succession of the Izumi basin offers an important frame of reference for studying the development of southwest Japan by the collision-accretion process since the Upper Cretaceous.

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References

Bando, Y., and H. Hashimoto, Biostratigraphy and ammonite fauna of the Izumi Group (Late Cretaceous) in the Asan mountains (in Japanese with English abstract), Mem. Fac. Educ. Kagawa Univ., II, 34, 11-39, 1984.

Crowell, J. C., Origin of late Cenozoic basins in southern California, in Tectonics and Sedimentation, edited by W. R. Dickinson: Soc. Econ. Paleont. Miner. Tursa Spec. Pub., 22, 190-204, 1974.

Faure, M., and F. Lalevee, Bent structural trends of Japan: Flexural-slip folding related to the Neogene opening of the Sea of Japan, Geology, 15, 49-52, 1987.

Ichikawa, K., Geohistory of the Median Tectonic Line of southwest Japan, Mem. Geol. Soc. Jpn., no. 18, 187-212, 1980.

Ichikawa, K., T. Matsumoto, and M. Iwasaki, Evolution of the Japanese Islands (in Japanese), Kagaku, 42, 181-191, 1972.

Jones, D. L., A. Cox, P. Coney, and M. Beck, The growth of Western North America, Sci. Am., Nov., 50-64, 1982.

Kanmera, K., and M. Hashimoto, The Izumi belt and its lateral extension (in Japanese), in The Geology of Japan, edited by K. Kanmera, M. Hashimoto, and T. Matsuda: Earth Sciences, 15, Iwanami Shoten, Tokyo, 37-39, 1980.

Kent, D. V., and F. M. Gradstein, A Cretaceous and Jurassic geochronology, Geol. Soc. Am. Bull., 96, 1419-1427, 1985.

Kodama, K., Two different paleomagnetic directions from the Izumi Group in Shikoku, Southwest Japan, J. Geomag. Geoelectr., 38, 279-284, 1986.

Kodama, K., Paleomagnetic constraints on the evolution of the Upper Cretaceous Izumi Basin in Shikoku, southwest Japan, in Terrane Accretion and Orogenic Belts, edited by E. C. Leitch and E. Scheibner: Geodyn. Ser. 19, Am. Geophys. Union, 291-299, 1987.

Kodama, K., A. Taira, M. Okamura, and Y. Saito, Paleomagnetism of the Shimanto belt in Shikoku, Southwest Japan, in Accretion Tectonics in the Circum-Pacific Regions, edited by M. Hashimoto and S. Uyeda, Terra Scientific Pub., Tokyo, 231-241, 1983.

MacDonald, W. D., Net tectonic rotation, apparent tectonic rotation, and the structural tilt correction in paleomagnetic studies, J. Geophys. Res., 85, 3659-3669, 1980.

Matsumoto, T., and Y. Morozumi, Late Cretaceous ammonites from the Izumi mountains, southwest Japan, Bull. Osaka Mus. Nat. Hist., 33, 1-16, 1980.

Matsumoto, T., H. Hashimoto, and M. Furuichi, An interesting species of Baculites (Ammonoidea) from the Cretaceous Izumi Group of Shikoku, Proc. Jpn. Acad., ser. B, 56, 408-413, 1980.

McFadden, P. L., and D. L. Jones, The fold test in palaeomagnetism, Geophys. J. Roy. Astron. Soc., 67, 53-58, 1981.

Morozumi, Y., Late Cretaceous (Campanian and Maastrichtian) ammonites from Awaji Island, Southwest Japan, Bull. Osaka Mus. Nat. Hist., 39, 1-58, 1985.

Murakami, N. (editor), Mesozoic acid igneous activity in Japan, Mem. Geol. Soc. Jpn., no. 17, 302 pp., 1979.

Nishimura, T., Basin analysis of the Upper Cretaceous Izumi Group in western Shikoku, Japan (in Japanese with English abstract), J. Geol. Soc. Jpn., 90, 157-174, 1984.

Noda, M., and M. Tashiro, The fauna from Dogo-Himezuka, Matsuyama City and its stratigraphic significance (in Japanese), J. Geol. Soc. Jpn., 79, 493-495, 1973.

Nur, A., and Z. Ben-Avraham, Break-up and accretion tectonics, in Accretion Tectonics in the Circum-Pacific Regions, edited by M. Hashimoto and S. Uyeda, Terra Scientific Pub., Tokyo, 3-18, 1983.

Okamura, M., Y. Kagawa, and M. Tashiro, Geology and radiolarians of the Izumi Group in the eastern part of Matsuyama City, Ehime Prefecture (in Japanese with English abstract), Res. Rep. Kochi Univ., 32, 339-347, 1984.

Otofuji, Y., and T. Matsuda, Timing of rotational motion of southwest Japan inferred from paleomagnetism, Earth. Planet. Sci. Lett., 70, 373-382, 1984.

Otofuji, Y., T. Matsuda, and S. Nohda, Brief review of Miocene opening of the Japan Sea: Paleomagnetic evidence from the Japan arc, J. Geomag. Geoelectr., 38, 287-294, 1986.

Otsuki, K., Plate tectonics of eastern Eurasia in the light of fault system, Sci. Rep. Tohoku Univ., 55, no. 2, 141-251, 1985.

Steel, R. J., Devonian basins of western Norway-Sedimentary response to tectonism and to varying tectonic context, Tectonophysics, 36, 207-224, 1976.

Suyari, K., On the lithofacies and the correlation of the Izumi Group of the Asan mountain range, Shikoku, Sci. Rep. Tohoku Univ., ser. 2, Spec. vol. 6, 489-495, 1973.

Suyari, K., and H. Hashimoto, A radiolarian assemblage from the Izumi Group of Eastern Shikoku (in Japanese with English abstract), J. Sci. Tokushima Univ., 18, 103-127, 1985.

Taira, A., Y. Saito, and M. Hashimoto, The fundamental processes of geologic evolution of Japan (in Japanese), Kagaku, 51, 508-515, 1981.

Taira, A., H. Okada, J. H. M. Whitaker, and A. J. Smith, The Shimanto Belt of Japan: Cretaceous-lower Miocene active-margin sedimentation, in Trench-Forearc Geology: Sedimentation and Tectonics on modern and ancient active plate margins, edited by J. K.

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Leggett: Geol. Soc. London Spec. Pub., 10, 5-26, 1982.

Takahashi, J., On "The Median Tectonic Line" in Matsuyama City and the neighboring areas, Ehime Prefecture, Japan, Mem. Fac. Educ., ser. III, Nat. Sci., 6, 1-44, 1986.

Yamasaki, T., Radiolarian assemblages of the Izumi Group in Shikoku and western Awaji Island, Southwest Japan, J. Geol. Soc. Jpn., 93, 403-417, 1987.

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